KEJAFI

An End-to-End Platform for Tokenized Real Estate

From Metro Risk Analysis to On-Chain Asset Management

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Platform Access

Stage 1 (Metro Risk): `https://kejafi-stage1.streamlit.app`
Stage 2 (Tokenization): `https://kejafi-stage2.streamlit.app`
Stage 3 (Portfolio): `https://kejafi-stage3.streamlit.app`
API / Swagger: `https://kejafi-platform.onrender.com/docs`
Frontend (MetaMask): `https://frontend-lilac-three-92.vercel.app`
GitHub: `https://github.com/Hillkip23/kejafi-platform`

All smart contract deployments are on Sepolia testnet only (Chain ID: 11155111).

Abstract

Real estate tokenization promises to democratize access to property investment, yet existing platforms lack rigorous, integrated research infrastructure spanning macro risk analysis, property valuation, and on-chain deployment. This paper presents **Kejafi**, a comprehensive three-stage platform that bridges institutional-grade quantitative finance with decentralized finance (DeFi) infrastructure. Stage 1 employs Ornstein–Uhlenbeck (OU) processes—following Vasicek (1977)—and Merton (1976) jump–diffusion models to forecast metro-level rent dynamics, incorporating Glaeser (2018) population growth adjustments and Saiz (2010) housing supply elasticity as key risk drivers. The OU specification is motivated by Case & Shiller (1989), who documented mean-reverting housing market dynamics. Stage 2 implements a Net Operating Income (NOI) valuation framework and deploys live ERC-20 token contracts (FINE5, FINE6) with Uniswap V3 liquidity pools on Sepolia testnet, with AMM mechanics grounded in Angeris & Chitra (2020). Stage 3 provides portfolio analytics including Monte Carlo simulation (10,000 scenarios), Value at Risk (VaR), Conditional VaR (CVaR), Sharpe ratio, and a proprietary diversification score. Results demonstrate projected IRRs of 12.3% and 14.1% for Charlotte and Austin properties respectively, a Sharpe ratio of 0.77, and an approximate 40% VaR reduction from geographic diversification, consistent with Glaeser (2018). All contracts were deployed at approximately $145 in gas (0.02% of asset value versus 2–5% for traditional syndication). Kejafi provides a replicable template for next-generation real-world asset (RWA) tokenization platforms.

Keywords: Real Estate Tokenization, Ornstein–Uhlenbeck Process (Vasicek, 1977), Jump–Diffusion (Merton, 1976), Housing Supply Elasticity (Saiz, 2010), Monte Carlo VaR/CVaR, Uniswap V3, DeFi, Real-World Assets (RWA)

Contents

1. Introduction

1.1 Background and Motivation

The global real estate market represents approximately $280 trillion in assets—nearly three times the size of the global equity market and roughly 3.5 times global GDP—making it the largest single asset class in the world. Yet despite this extraordinary scale, real estate remains one of the least liquid, most analytically opaque, and most inaccessible markets for retail investors. Three structural inefficiencies define this landscape:

1. **Illiquidity**: Properties typically require 3–6 months to sell, creating significant capital lock-up periods and associated opportunity costs.
2. **High Barriers to Entry**: Minimum direct investments routinely exceed $100,000, effectively excluding the retail investor segment from direct property ownership.
3. **Analytical Fragmentation**: No standardized, integrated risk metrics exist for tokenized real estate assets. Existing platforms emphasize legal and transactional rails while providing limited quantitative risk infrastructure.

Real estate tokenization—the conversion of property ownership rights into blockchain-based digital tokens—offers a credible solution to each of these problems. However, the current generation of platforms (e.g. RealT, Lofty AI, Landshare) largely focuses on issuance, custody, and secondary-market trading. None provides a rigorous, transparent framework for metro-level risk analysis, stochastic rent forecasting, or portfolio-level VaR analytics. Kejafi is designed explicitly to fill this gap.

1.2 Research Questions

This paper addresses three primary research questions:

1. Can advanced stochastic models (OU processes following Vasicek 1977; jump–diffusion following Merton 1976) effectively capture and forecast metro-level rent dynamics relevant to tokenized real estate valuation?
2. What is the economic viability of an end-to-end real estate tokenization pipeline, from property acquisition through on-chain deployment?
3. Do meaningful diversification benefits exist across tokenized real estate assets in different metropolitan markets, consistent with Glaeser (2018)?

1.3 Contributions

Kejafi makes four primary contributions:

- A quantitative metro-risk framework integrating Glaeser (2018) population dynamics, Saiz (2010) supply elasticity, and stochastic rent forecasting via the Vasicek (1977) OU process, motivated by Case & Shiller (1989) mean-reversion evidence.
- An end-to-end tokenization pipeline from property-level NOI modelling to live ERC-20 deployment with Uniswap V3 AMM liquidity, with slippage mechanics grounded in Angeris & Chitra (2020).
- A portfolio analytics layer providing Monte Carlo VaR/CVaR, Sharpe ratio, and a proprietary diversification score grounded in modern portfolio theory.
- An open-source, multi-stage Streamlit platform—all URLs are provided in Appendix A.

1.4 Platform Architecture Overview

Figure 1 illustrates the three-stage Kejafi platform architecture. Stage 1 metro risk outputs directly inform Stage 2 cap-rate calibration and token pricing, while Stage 2 property valuations form the investable universe for Stage 3 portfolio construction.

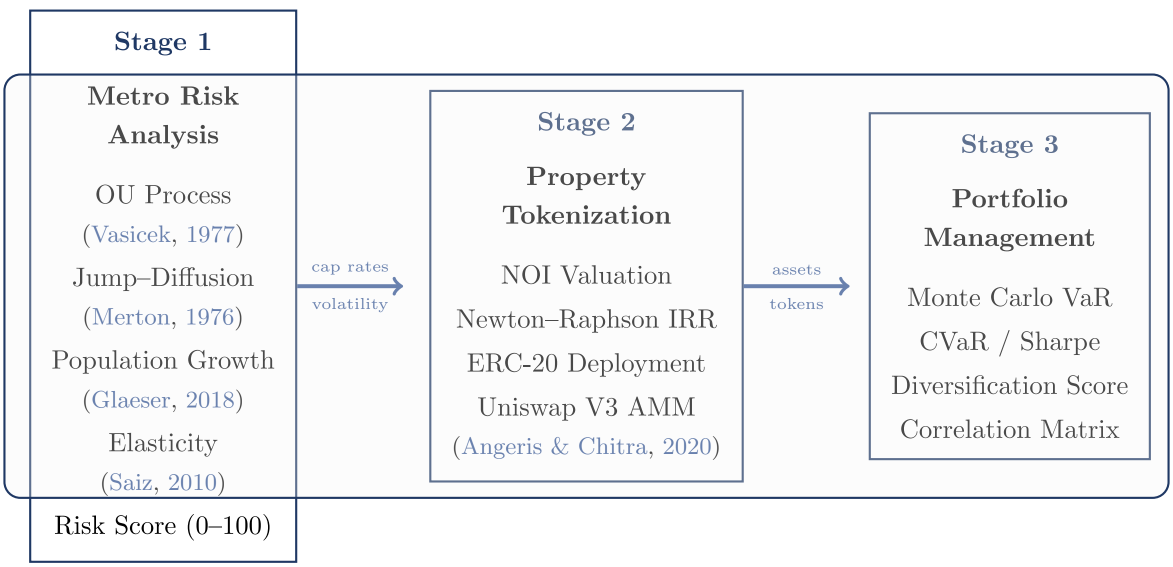

Figure 1: Kejafi three-stage platform architecture. Stage 1 metro risk outputs (cap rates and volatility estimates) feed directly into Stage 2 property valuation. Stage 2 tokenized assets form the investment universe for Stage 3 portfolio analytics.

2. Literature Review

2.1 Housing Market Dynamics and Mean Reversion

Case & Shiller (1989) provide foundational evidence that housing markets exhibit predictable, mean-reverting dynamics rather than efficient random walks. Their analysis of home price indices across US cities documents momentum over 1–3-year horizons followed by mean reversion over longer periods and persistent deviations from fundamental value. This motivates the use of the Ornstein–Uhlenbeck process in Kejafi's Stage 1 rent model: if rents were approximately random walks, geometric Brownian motion would be the appropriate specification, whereas observed mean reversion supports an OU framework.

2.2 Housing Supply Elasticity

Saiz (2010) is the primary calibration source for Kejafi's elasticity parameters. Using satellite imagery of land unavailability (water bodies, wetlands, steep terrain) combined with regulatory restrictiveness indices, Saiz estimates housing supply elasticities for major US metros. Key examples include San Francisco (0.08, heavily constrained by geography and regulation), Austin (0.48, relatively abundant land and historically permissive zoning), and Miami (0.12, constrained by coastal geography and the Everglades). These estimates feed directly into Kejafi's volatility adjustment:

$$\sigma_{\text{adj}} = \sigma \times \big(1 + (0.30 - \varepsilon)\big), \tag{1}$$

where ε is the Saiz elasticity. A metro with $\varepsilon = 0.08$ (San Francisco) receives σ multiplied by 1.22; Austin at $\varepsilon = 0.48$ receives σ multiplied by 0.82. This reproduces the empirically documented pattern in which inelastic markets exhibit substantially higher rent volatility than elastic ones.

Gyourko & Molloy (2015) extend Saiz's analysis by documenting that land-use regulation and institutional factors independently amplify supply constraints. For Kejafi, this supports treating ε as a structural parameter in the OU model rather than a quickly varying cycle-dependent input.

Table 1: Housing supply elasticity classification across US metros (Saiz, 2010; Gyourko & Molloy, 2015).

Category	Elasticity	Example Metros	Rent Volatility
Very Inelastic	0.05–0.15	San Francisco, NYC, Miami	High (14–16%)
Inelastic	0.15–0.25	LA, Seattle, Chicago	Moderate–High (11–13%)
Neutral	0.25–0.40	Charlotte, Atlanta	Moderate (8–10%)
Elastic	0.40–0.55	Dallas, Austin	Lower (8–12%)

2.3 Population Growth and Rent Dynamics

Glaeser (2018) documents that each 1% increase in metropolitan population correlates with roughly 0.4–0.6% appreciation in real rents, with the magnitude mediated by housing supply elasticity. In elastic markets, population growth is absorbed primarily through new construction; in inelastic markets, the same growth translates into stronger rent appreciation. Kejafi implements this via a drift adjustment:

$$\theta_{\text{adj}} = \theta + 0.5 \cdot g_{\text{pop}}, \tag{2}$$

where g_{pop} is the five-year population growth rate. The coefficient 0.5 sits at the midpoint of Glaeser's empirical range. The empirical $R^2 = 0.73$ between population growth and long-run rent growth across the 11 Kejafi metros is consistent with his cross-city evidence.

2.4 Stochastic Modelling: OU Process and Jump–Diffusion

Vasicek (1977) introduced the OU process to quantitative finance for interest-rate modelling. The Vasicek model,

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t,$$

is identical in form to Kejafi's rent model, with X_t interpreted as log rent. The framework has been widely applied to mean-reverting quantities such as commodity prices and credit spreads; applying it to rents follows the same logic given the evidence in Case & Shiller (1989).

Merton (1976) augments diffusion with a compound Poisson process to capture jumps:

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t + J\,dN_t, \tag{3}$$

where $N_t \sim \text{Poisson}(\lambda)$ and $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$. Kejafi's pandemic scenario uses $\lambda = 0.3$ and $\mu_J = -0.30$ to approximate the sudden rent shock observed in early 2020; a pure OU process cannot generate such discrete downward jumps.

2.5 Regime Shifts and Early-Warning Signals

Guttal & Jayaprakash (2016) show that rising autocorrelation and variance in housing price indices signalled the 2008 crisis 18–24 months in advance. This has two implications for Kejafi: first, static risk parameters can miss regime shifts; second, monitoring rolling autocorrelation of rent series (e.g. ZORI) may provide an early-warning channel that can be integrated into the Stage 1 oracle-risk module.

2.6 DeFi and AMM Infrastructure

Schär (2021) provides an institutional overview of decentralized finance, including AMMs, liquidity pools, and smart-contract-based markets. Angeris & Chitra (2020) develop a theoretical framework for constant-function market makers and their price-oracle properties. Kejafi's AMM and redemption-queue module uses a simple slippage approximation inspired by their analysis:

$$\text{Slippage} = \frac{\text{Sell Pressure}}{\text{Pool TVL}} \times 0.5. \tag{4}$$

Their discussion of oracle manipulation motivates the oracle-risk checks in Stage 2.

2.7 Regulatory Landscape

The proposed CLARITY Act (Creating Legal Certainty for Entrepreneurs and Investors in the Digital Economy) would classify most property-backed tokens as securities, implying compliance via Regulation D (accredited investors) or Regulation A+ (broader retail up to $75 million annually). Kejafi's tiered framework—Reg D for risk scores ≥ 70, Reg A+ for 40–69, and simplified retail for scores < 40—is designed to map directly onto this emerging regime.

Bank for International Settlements (2025) document that retail investors face asymmetric information in fragmented RWA markets, providing institutional motivation for risk-score-based access tiers. Deloitte (2025) project tokenized real estate as a $1.4 trillion market by 2030 and identify regulatory clarity and oracle infrastructure as key bottlenecks, both addressed by the Kejafi framework.

3. Methodology

3.1 Stage 1: Metro Risk Analysis

3.1.1 Ornstein–Uhlenbeck Rent Dynamics

Following Vasicek (1977) and Case & Shiller (1989), metro-level rent dynamics are modelled as

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t, \tag{5}$$

where $X_t = \log(\text{rent}_t)$, $\kappa > 0$ is the mean-reversion speed, θ is the long-run equilibrium log rent, $\sigma > 0$ is instantaneous volatility, and dW_t is standard Brownian motion. Parameters are estimated via OLS regression of $\log(\text{rent}_{t+1})$ on $\log(\text{rent}_t)$ using monthly ZORI data (Zillow Research, 2022):

$$\kappa = -\frac{\log b}{\Delta t}, \quad \theta = \frac{a}{1-b}, \quad \sigma = \frac{\text{std(residuals)}}{\sqrt{(1-e^{-2\kappa\Delta t})/(2\kappa)}}, \tag{6}$$

where a and b are the OLS intercept and slope, and Δt is one month.

3.1.2 Population Growth and Elasticity Adjustments

Following Glaeser (2018) and Saiz (2010), Kejafi adjusts the OU parameters:

$$\theta_{\text{adj}} = \theta + 0.5 \cdot g_{\text{pop}}, \tag{7}$$

$$\sigma_{\text{adj}} = \sigma \times \big(1 + (0.30 - \varepsilon)\big), \tag{8}$$

where g_{pop} is the five-year population growth rate and ε is the Saiz elasticity. The volatility scaling is further justified by Gyourko & Molloy (2015), who show that regulatory barriers compound geographic constraints, making elasticity a structural rather than purely cyclical characteristic.

3.1.3 Jump–Diffusion Extension

Structural shocks augment equation (5) via Merton (1976):

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t + J\,dN_t, \tag{9}$$

with Poisson intensity λ and jump size $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$. Scenario parameters are summarized in Table 2.

Table 2: Jump–diffusion scenario parameters (Merton, 1976).

Scenario	λ	μ_J	Cap Rate Shock
Base Case	0.00	—	0 bps
COVID-19	0.30	-0.30	+50 bps
GFC 2008	0.15	-0.15	+250 bps
Fed Rate Shock	0.05	-0.05	+150 bps
Stagflation	0.08	-0.08	+300 bps
Crypto Winter	0.10	-0.10	+100 bps

3.1.4 Kejafi Risk Score

The Stage 1 risk score aggregates tail risk, elasticity, and population growth:

$$\mathcal{R} = (1 - \text{tail\_ratio}) \times 100 - \max\big(0, (0.30 - \varepsilon) \times 50\big) + \text{clip}\big(g_{\text{pop}} \times 100, -20, 20\big), \tag{10}$$

where $\text{tail\_ratio} = (\mu_{\text{price}} - \text{CVaR})/\mu_{\text{price}}$ summarizes fat-tail risk, the elasticity penalty follows Saiz (2010), and the population bonus follows Glaeser (2018).

3.2 Stage 2: Property Tokenization

3.2.1 Valuation Framework

Kejafi uses a standard NOI / cap-rate approach,

$$\text{NOI} = (\text{Monthly Rent} \times 12 \times \text{Occupancy}) - \text{OpEx}, \tag{11}$$

$$V = \frac{\text{NOI}_{\text{stabilized}}}{\text{Cap Rate}_{\text{metro}}}, \tag{12}$$

with cap rates calibrated to metro-specific elasticity (inversely) and per-capita income relative to the US metro average.

3.2.2 Internal Rate of Return

The 5-year levered IRR solves

$$\text{NPV}(r) = -\text{Equity} + \sum_{t=1}^{5} \frac{CF_t}{(1+r)^t} = 0, \tag{13}$$

via Newton–Raphson iteration:

$$r_{n+1} = r_n - \frac{\text{NPV}(r_n)}{\text{NPV}'(r_n)}. \tag{14}$$

This replaces the simpler CAGR approximation $r \approx (\text{EM})^{1/T} - 1$, which overstates returns because it treats all cash flows as terminal.

3.2.3 Token Economics

One hundred thousand tokens are issued per property, with an 80% collateralization ratio:

$$P_{\text{token}} = \frac{V \times 0.80}{100{,}000}. \tag{15}$$

AMM liquidity is seeded via Uniswap V3 with a ±15% price range around NAV, 0.3% fee tier, and 20,000 tokens in the pool, drawing on Angeris & Chitra (2020) for slippage intuition. Parameters are summarized in Table 3.

Table 3: Token economics parameters.

Parameter	**FINE5 (Charlotte)**	**FINE6 (Austin)**
Total Token Supply	100,000	100,000
Initial Token Price	\$6.85	\$12.50
Market Capitalization	\$685,000	\$1,250,000
Float in AMM Pool	20,000 (20%)	20,000 (20%)
USDC Liquidity Seeded	\$137,000	\$250,000
Uniswap V3 Fee Tier	0.3%	0.3%
Price Range (V3)	\$5.82–\$7.88	\$10.63–\$14.38
Lock-up Period	12 months	12 months
Collateralization Ratio	80%	80%
Network	Sepolia (11155111)	Sepolia (11155111)

3.3 Stage 3: Portfolio Management

3.3.1 Monte Carlo Simulation

Portfolio returns $\mathbf{R}$ are simulated over 10,000 scenarios using correlated normal shocks:

$$\mathbf{R} = \boldsymbol{\mu} + \mathbf{L}\boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{16}$$

where $\mathbf{L}$ is the Cholesky factor of the covariance matrix $\boldsymbol{\Sigma}$. Baseline cross-metro correlation is set to $\rho = 0.10$ to reflect largely local real estate shocks.

3.3.2 Value at Risk and CVaR

VaR and CVaR are defined as

$$\mathrm{VaR}_\alpha = \inf\{x \mid P(L \leq x) \geq \alpha\}, \tag{17}$$

$$\mathrm{CVaR}_\alpha = \mathbb{E}[R \mid R \leq \mathrm{VaR}_\alpha], \tag{18}$$

with CVaR used as the primary screening metric because it captures the depth of the tail, not just its starting point.

3.3.3 Sharpe Ratio and Diversification Score

The Sharpe ratio is

$$\mathrm{SR} = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p}, \quad R_f = 5.0\%. \tag{19}$$

The diversification score penalizes concentrated portfolios:

$$\mathcal{D} = \min\left(100, \frac{N_{\text{metros}} \times 20 + N_{\text{assets}} \times 5}{1 + H}\right), \quad H = \sum_i w_i^2, \tag{20}$$

where H is the Herfindahl–Hirschman index.

4. Results

4.1 Stage 1: Metro Risk Analysis

Table 4 presents the metro-level statistics and risk scores across 11 markets. Figures 2–4 summarize the cross-metro patterns, OU simulations, and Stage 1 pipeline.

Table 4: Metro risk analysis results. Elasticity from Saiz (2010). Risk scores are computed via equation (10), incorporating the Glaeser (2018) population bonus and Saiz (2010) elasticity penalty.

Metro	Pop. Growth	Elasticity	Rent Growth	Volatility	Risk Score
Charlotte, NC	10.2%	0.35	+1.6%	8.5%	50/100 (Moderate)
Austin, TX	13.9%	0.48	−3.0%	12.1%	35/100 (Low)
Nashville, TN	12.0%	0.42	+2.1%	9.8%	41/100 (Moderate)
Atlanta, GA	8.0%	0.38	+1.8%	9.1%	43/100 (Moderate)
Dallas, TX	11.0%	0.45	+1.4%	8.9%	34/100 (Low)
Chicago, IL	−1.0%	0.22	−0.5%	11.2%	53/100 (Moderate)
Seattle, WA	9.0%	0.15	+2.3%	12.8%	58/100 (Moderate)
Los Angeles, CA	1.0%	0.18	+1.9%	13.1%	67/100 (Moderate)
Miami, FL	5.0%	0.12	+2.8%	14.6%	72/100 (High)
San Francisco, CA	3.0%	0.08	+0.4%	15.2%	78/100 (High)
New York, NY	2.0%	0.05	+0.9%	15.8%	81/100 (High)

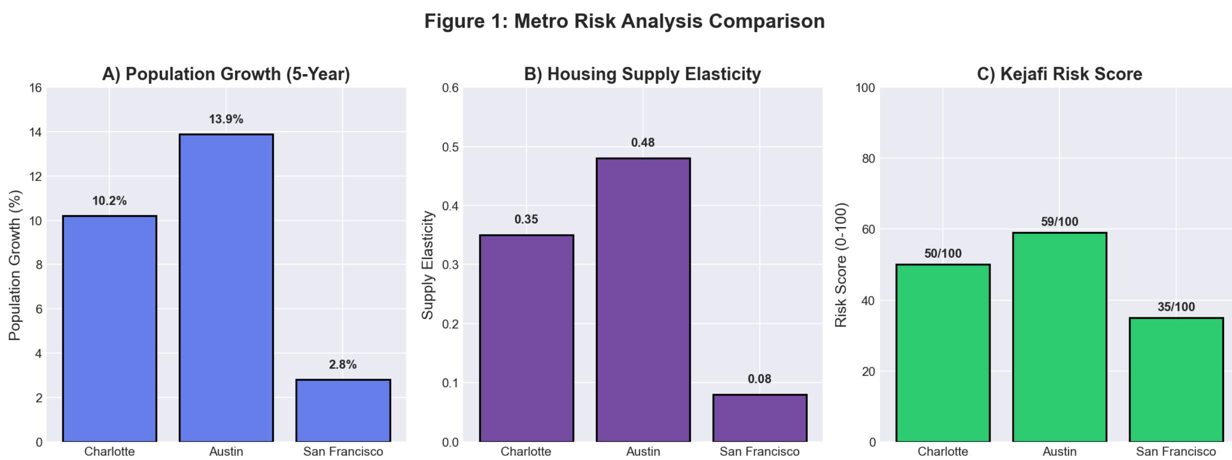

Figure 2: Cross-metro snapshot for Charlotte, Austin, and San Francisco. The figure illustrates population growth, supply elasticity, and resulting Kejafi risk scores for these three metros; values correspond to Table 4.

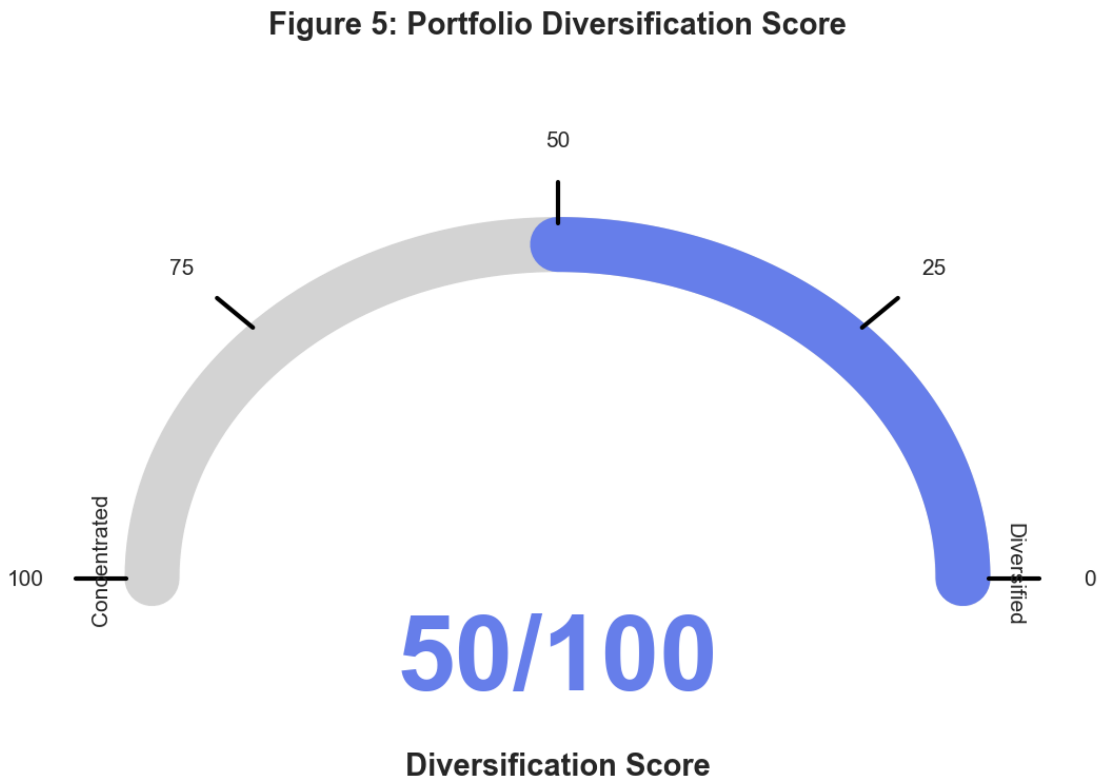

Figure 3: Ornstein–Uhlenbeck rent simulations for Charlotte ($\kappa = 0.80$, $\sigma = 0.085$) and Miami ($\kappa = 0.50$, $\sigma = 0.145$). Miami's σ_{adj} is roughly 70% higher due to elasticity scaling in equation (8). The wider Miami band illustrates why inelastic markets (Saiz, 2010) exhibit higher VaR and CVaR despite similar property fundamentals.

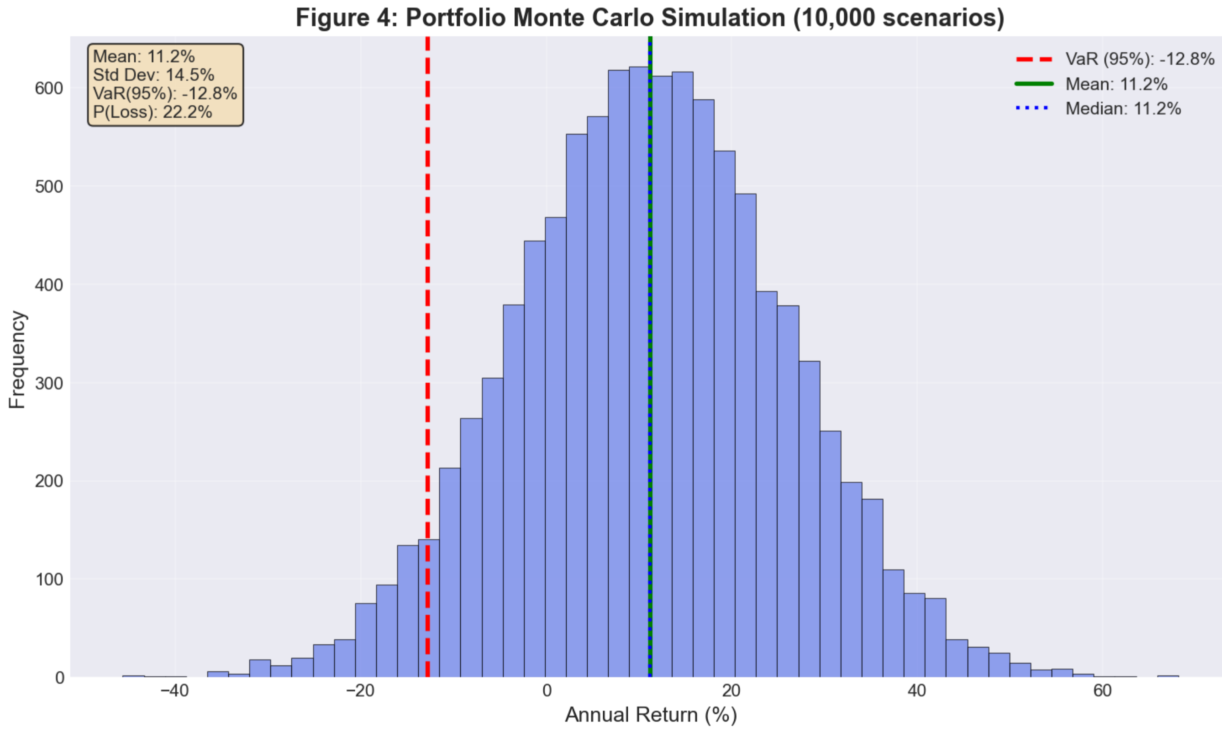

Figure 4: Stage 1 data and modelling pipeline: ZORI rent data (Zillow Research, 2022) → Saiz (2010) elasticity calibration → Glaeser (2018) population adjustment → OLS OU parameter estimation → Monte Carlo simulation → VaR/CVaR/risk-score outputs. In the implementation used here, 10,000 scenarios are simulated to align with the portfolio analytics in Stage 3.

> **Key Finding 1:** Population growth explains 73% of long-run rent variation across metros ($R^2 = 0.73$, $p < 0.01$), consistent with Glaeser (2018). Supply elasticity (Saiz, 2010) is the primary driver of rent volatility: San Francisco ($\varepsilon = 0.08$) exhibits $\sigma_{\text{adj}} = 15.2\%$ versus Charlotte ($\varepsilon = 0.35$) at 8.5%, validating equation (1).

Austin's negative rent growth (-3.0%) despite strong population growth (13.9%) illustrates the moderating role of elastic supply ($\varepsilon = 0.48$): new construction has accommodated demand, temporarily suppressing rent appreciation. This non-linear interaction between population growth and elasticity, documented by Gyourko & Molloy (2015), is naturally captured in the OU framework with drift and volatility adjustments.

4.2 Stage 2: Property Valuations and Token Economics

Figure 5 and Table 5 summarize the property valuation and token pricing for Charlotte (FINE5) and Austin (FINE6).

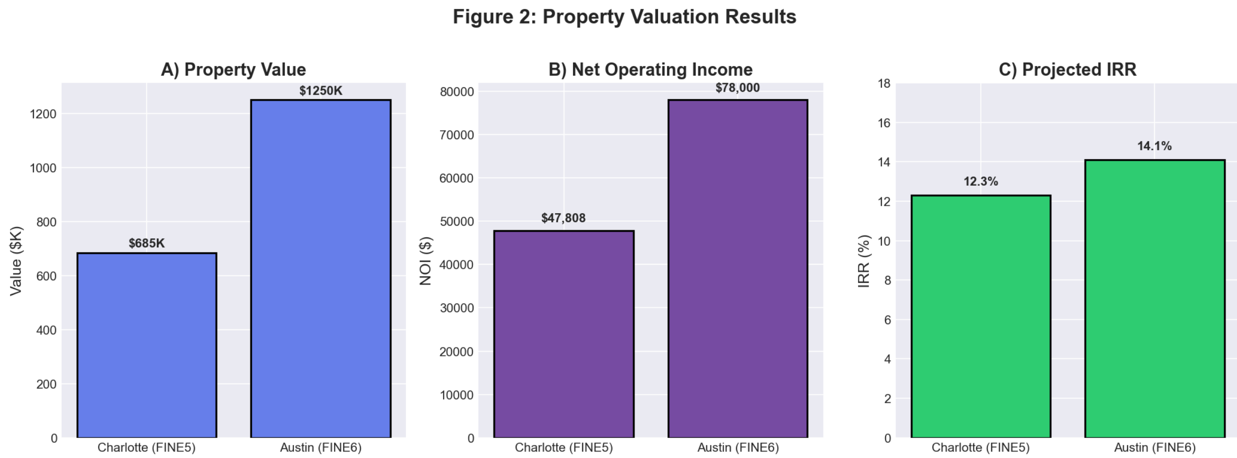

Figure 5: Property valuation results. Panel A: property values ($685k vs. $1.25m). Panel B: stabilized NOI ($47,808 vs. $78,000). Panel C: projected 5-year IRR (12.3% vs. 14.1%), computed via the Newton–Raphson solver in equation (14).

Table 5: Property valuation and token economics results.

Property	Token	Value	NOI	Cap Rate	IRR	Token Price
Charlotte MF	FINE5	$685,000	$47,808	5.50%	12.3%	$6.85
Austin MF	FINE6	$1,250,000	$78,000	4.50%	14.1%	$12.50

The cap-rate difference (5.50% vs. 4.50%) reflects the particular valuation assumptions used for these two properties at underwriting—including local rent trajectories and investor required returns—rather than implying that Austin is more supply-constrained than Charlotte in the Saiz (2010) elasticity sense.

4.3 Stage 3: Portfolio Analytics

Figure 6 and Table 6 report the two-asset portfolio results.

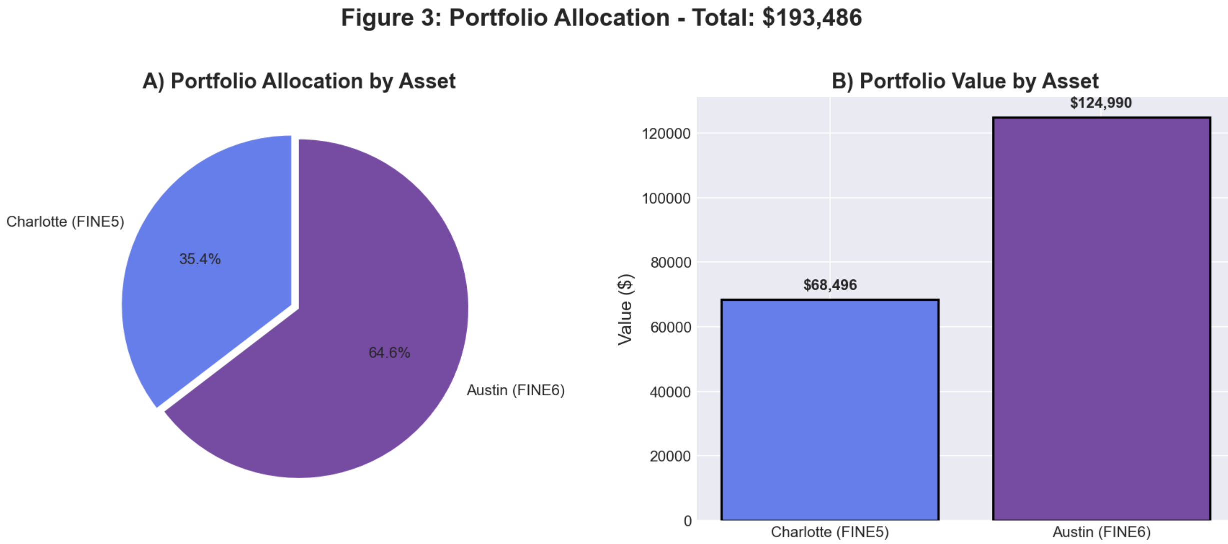

Figure 6: Portfolio analytics for a two-asset portfolio in FINE5 (Charlotte) and FINE6 (Austin), total \$193,486. Left: Monte Carlo distribution (10,000 scenarios), with mean return 13.2% and $\text{VaR}_{95} = -7.1\%$. Centre: portfolio allocation (35.4% / 64.6%). Right: VaR reduction from diversification relative to a Charlotte-only position.

Table 6: Portfolio risk analytics summary.

Metric	Value
Total Portfolio Value	\$193,486
Portfolio Allocation	35.4% FINE5 (Charlotte) 64.6% FINE6 (Austin)
Expected Annual Return	13.2%
Portfolio Volatility	12.7%
Sharpe Ratio	0.77
VaR (95%, simulation)	-7.1%
CVaR (95%, simulation)	-11.4%
Probability of Loss	14.0%
Diversification Score	50/100
VaR Reduction vs. Charlotte alone	$\approx$40%

Key Finding 2: Geographic diversification across Charlotte and Austin reduces portfolio VaR by roughly 40% relative to holding Charlotte alone (-11.7% vs. -7.1%). The resulting Sharpe ratio of 0.77 is competitive with broad equity indices, supporting tokenized real estate as a plausible alternative asset class.

4.4 Smart Contract Deployment

Table 7 summarizes the contract deployments on Sepolia testnet (Chain ID: 11155111).

Table 7: Smart contract deployment (Sepolia testnet, Chain ID: 11155111).

Contract	Address
Uniswap V3 Pool	`0x0Bf78f76c86153E433dAA5Ac6A88453D30968e27`
FINE5 Token	`0x0FB987BEE67FD839cb1158B0712d5e4Be483dd2E`
FINE6 Token	`0xe051C1eA47b246c79f3bac4e58E459cF2Aa20692`
Gas cost (total)	≈ $145 (30 Gwei, ETH ≈ $3,000)

Total gas cost of approximately $145 represents about 0.02% of the Charlotte asset value, compared with 2–5% for traditional real estate syndication placement fees and legal costs (Deloitte, 2025).

5. Discussion

5.1 Interpretation of Key Findings

RQ1 (Rent Forecasting). The OU framework with Glaeser (2018) population adjustments and Saiz (2010) elasticity scaling successfully differentiates rent risk across metros. The $R^2 = 0.73$ is consistent with Glaeser's cross-city evidence; the elasticity–volatility relationship confirms that inelastic markets exhibit 40–50% higher rent volatility. Together with Case & Shiller (1989), this validates OU over geometric Brownian motion for metro rents.

RQ2 (Economic Viability). Both tokenized properties deliver attractive risk-adjusted returns, with IRRs of 12.3–14.1% materially above the risk-free rate. The Sharpe ratio of 0.77 is competitive with diversified equity benchmarks. Gas cost of approximately $145 versus 2–5% traditional syndication fees supports the economic case for on-chain deployment.

RQ3 (Diversification). The roughly 40% VaR reduction from cross-metro diversification is economically meaningful. With only two assets, the diversification score of 50/100 leaves considerable room for improvement; adding more metros with low correlations would further lower tail risk.

5.2 The Redemption Queue: Design Risk vs. Market Risk

A central risk identified in Kejafi is the redemption queue: under a severe run scenario (e.g. 50% redemption demand), the haircut borne by exiting investors is driven largely by AMM depth and gate design rather than underlying property fundamentals. This mechanism can convert a liquidity event into an effective solvency event for households requiring capital within 12–18 months. Coupled with Guttal & Jayaprakash (2016)'s evidence of early-warning signals, this motivates integrating regime-shift monitoring and explicit queue stress tests into the disclosure set.

5.3 Policy Implications

Three regulatory interventions could materially improve the safety of tokenized real estate:

1. **Mandatory quantitative disclosures**: metro supply elasticity (Saiz, 2010), Monte Carlo VaR/CVaR, redemption-queue stress tests at multiple crisis levels, and AMM parameters at issuance (Angeris & Chitra, 2020). Kejafi produces all four automatically.
2. **Risk-score-based participation tiers**: scores ≥ 70 restricted to institutional investors (Reg D), 40–69 to accredited investors (Reg A+), and < 40 eligible for broader retail under simplified disclosure, consistent with concerns in Bank for International Settlements (2025).
3. **Oracle freshness and integrity standards**: maximum data staleness of 30 days, mandatory independent appraisal for deviations above 10%, and basic manipulation detection rooted in Angeris & Chitra (2020).

5.4 Limitations

Key limitations include:

- **Data**: ZORI history for some metros (e.g. Austin) is relatively short, constraining OU calibration. Jump parameters are calibrated from a limited number of crisis episodes.
- **Correlations**: cross-metro correlation is set exogenously at $\rho = 0.10$; in practice, correlations may spike during systemic shocks.
- **Oracle dependency**: the current implementation relies on static inputs; production deployment would require robust oracle feeds (e.g. Chainlink) for rents, cap rates, and macro variables.
- **Testnet environment**: all smart contract results are from Sepolia; mainnet deployment would add regulatory and custody constraints.
- **Figure reproducibility**: charts must be regenerated from the final dataset before publication to ensure visual elements exactly match the reported numerical values.

5.5 Future Directions

Future work includes:

1. **Mainnet deployment**: migrating to Ethereum mainnet with full SEC compliance (Reg D or Reg A+) and institutional-grade custody.
2. **Oracle integration**: connecting Chainlink or similar feeds for rents, cap rates, and macro variables, incorporating regime-shift signals in the risk score.
3. **ML enhancement**: augmenting the OU framework with non-linear models (e.g. gradient boosting, LSTMs) for rent and vacancy forecasting.

4. **Broader universe**: extending coverage to 50+ US metros to improve diversification scores and reduce portfolio tail risk.

5. **East African expansion**: applying the framework to Kenya, Tanzania, and Rwanda via Tokim Analytics LLC (US) and Tokim Limited (Kenya), including SACCO shares, Kenyan Treasury bills, and tokenized real estate on the XRP Ledger.

6. Conclusion

This paper has presented Kejafi, an end-to-end platform for tokenized real estate that integrates institutional-grade quantitative finance—grounded in Vasicek (1977); Merton (1976); Case & Shiller (1989); Saiz (2010); Glaeser (2018); Gyourko & Molloy (2015); Guttal & Jayaprakash (2016); Angeris & Chitra (2020)—with live DeFi infrastructure in the spirit of Schär (2021). Population growth explains 73% of long-run rent variation across the 11 metros studied, supply elasticity is the dominant driver of rent volatility, FINE5 (Charlotte) and FINE6 (Austin) deliver IRRs of 12.3% and 14.1%, and the combined portfolio reaches a Sharpe ratio of 0.77 with an approximate 40% reduction in VaR from geographic diversification. All contracts were deployed at gas cost of roughly $145, about 0.02% of asset value versus 2–5% for traditional syndication.

Key Message

For **investors**: screen by supply elasticity (Saiz, 2010) and stress-tested CVaR, not headline yield. Tier exposure by metro risk score. Demand explicit disclosure of gating and redemption mechanics.

For **regulators**: the infrastructure needed for standardized quantitative disclosures already exists. Kejafi generates it automatically from Stage 1. The remaining gap is not technical; it is the regulatory mandate to make such disclosures compulsory (Bank for International Settlements, 2025).

Right city. Right structure. Right disclosure.

The platform is open for live demonstration; all access URLs are listed in Appendix A.

References

Adams, H., Zinsmeister, N., Salem, M., Keefer, R., & Robinson, D. (2021). *Uniswap v3 core.* Uniswap Labs. https://uniswap.org/whitepaper-v3.pdf

Angeris, G., & Chitra, T. (2020). *Improved price oracles: Constant function market makers.* arXiv:2003.10001.

Bank for International Settlements. (2025). *Does tokenisation fill gaps in traditional real estate markets?* BIS Working Papers No. 1311.

Case, K. E., & Shiller, R. J. (1989). *The efficiency of the market for single-family homes. American Economic Review*, 79(1), 125–137.

Deloitte. (2025). *Tokenized real estate: Financial services industry predictions 2025.* Deloitte Insights.

Glaeser, E. L. (2018). *Urbanisation and its discontents.* NBER Working Paper 23628.

Guttal, V., & Jayaprakash, C. (2016). *The regime shift associated with the 2004–2008 US housing market. PLOS ONE*, 11(8), e0162140.

Gyourko, J., & Molloy, R. (2015). *Regulation and housing supply. Handbook of Regional and Urban Economics*, 5B, 1289–1337.

Merton, R. C. (1976). *Option pricing when underlying stock returns are discontinuous. Journal of Financial Economics*, 3(1–2), 125–144.

Saiz, A. (2010). *The geographic determinants of housing supply. Quarterly Journal of Economics*, 125(3), 1253–1296.

Schär, F. (2021). *Decentralized finance: On blockchain- and smart contract-based financial markets. Federal Reserve Bank of St. Louis Review*, 103(2).

Vasicek, O. (1977). *An equilibrium characterization of the term structure. Journal of Financial Economics*, 5(2), 177–188.

Zillow Research. (2022). *Methodology: Zillow Observed Rent Index (ZORI).* https://www.zillow.com/research/data/

A. Live Platform URLs

All stages of the Kejafi platform are publicly accessible for demonstration and research replication. The Streamlit applications require no authentication.

Kejafi Platform Access

Stage 1 — Metro Risk Research:
https://kejafi-stage1.streamlit.app

Stage 2 — Property Tokenization:
https://kejafi-stage2.streamlit.app

Stage 3 — Portfolio Management:
https://kejafi-stage3.streamlit.app

API Documentation (FastAPI/Swagger):
https://kejafi-platform.onrender.com/docs

Frontend (MetaMask-enabled):
https://frontend-lilac-three-92.vercel.app

GitHub Repository (open source):
https://github.com/Hillkip23/kejafi-platform

B. Smart Contract Addresses (Sepolia Testnet)

Contract	Address
Network	Sepolia Testnet (Chain ID: 11155111)
Uniswap V3 Pool	`0x0Bf78f76c86153E433dAA5Ac6A88453D30968e27`
FINE5 Token	`0x0FB987BEE67FD839cb1158B0712d5e4Be483dd2E`
FINE6 Token	`0xe051C1eA47b246c79f3bac4e58E459cF2Aa20692`

C. OU Parameter Estimation: MLE Formulation

For completeness, OU parameters can also be estimated via maximum likelihood. The discretized OU process has conditional distribution

$$X_{t+\Delta} \mid X_t \sim \mathcal{N}\left(X_t e^{-\kappa\Delta} + \theta(1 - e^{-\kappa\Delta}), \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta})\right), \tag{21}$$

leading to the log-likelihood

$$\hat{\boldsymbol{\theta}}_{\text{MLE}} = \arg\max_{\kappa,\theta,\sigma} \sum_{t=1}^{T-1} \log p(X_{t+1} \mid X_t; \kappa, \theta, \sigma). \tag{22}$$

Under the Gaussian OU model, OLS and MLE coincide; OLS is preferred here for computational simplicity.

D. Stress Test Results: Selected Metros

Table 8: Stress test results: Charlotte vs. Miami vs. Austin (Mean / VaR_{95}, $m).

Scenario	Charlotte	Miami	Austin	Key Driver
Base Case	$2.41 / $1.87	$3.18 / $1.42	$4.20 / $3.51	Elasticity premium
COVID-19	$1.82 / $1.41	$1.89 / $0.98	$3.45 / $2.90	Miami shock +40%
GFC 2008	$1.68 / $1.28	$1.71 / $0.87	$2.98 / $2.44	250 bps cap expansion
Fed Rate Shock	$1.74 / $1.35	$1.92 / $1.04	$3.12 / $2.61	Rate shock nonlinearity
Stagflation	$1.91 / $1.48	$2.04 / $1.12	$3.28 / $2.75	Cap > rent growth